

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected December 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C/C++, Python, Assembly, Java

Electric Powertrain Controls: Fixed-point PI/PID control, power limiting, energy management, saturation, anti-windup, slew-rate limiting

Embedded Systems: STM32H755, TTC 60 VCU, PX4, real-time firmware, CAN, SPI, microcontroller I/O

Test & Tools: Software-in-the-loop, CAN trace analysis, TASKING, BusMaster, MoTeC i2 Pro, GDB, Git

EXPERIENCE

Software Lead

Jun 2026 to Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead the software organization for a Formula SAE electric racecar, directing **12 engineers** across **10+ concurrent projects** in four-motor controls, BMS and dashboard firmware, thermal derating, plant modeling, and telemetry
- Direct four-motor controls and validation, defining a phased torque vectoring and traction control program and coordinating battery, motor, and inverter plant models for software- and hardware-in-the-loop testing

Software Engineer Intern

Feb 2026 to Present

Argus Defense | San Jose, CA

- Tuned PID flight-control loops on a PX4 flight controller to reduce vertical and horizontal position drift in autonomous flight
- Integrated and tuned visual-inertial odometry for onboard state estimation, improving position-hold accuracy
- Developed and validated embedded flight-control firmware in C on PX4, iterating control gains against logged flight telemetry

Software Controls Engineer

Jul 2025 to Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Designed a lap-adaptive energy-management algorithm that sets a power setpoint from cumulative energy versus budget, with measured power tracking the commanded setpoint within **0.01%**
- Developed embedded power-limiting firmware with a PI controller on the VCU, maximizing motor output while enforcing the Formula SAE **80 kW** limit
- Built CAN diagnostics firmware for live power and torque frames, status bits, and error counters for real-time BusMaster monitoring

Software Intern

Aug 2024 to Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Built deterministic entry and exit state logic for power limiting from measured power, requested torque, and threshold flags
- Tuned the PI torque controller used for power limiting to within **2%** error, smoothing torque transitions to preserve drivability
- **Results:** 1st in endurance at MIS EV 2025

PROJECT EXPERIENCE

Formula SAE EV Vehicle Simulation

Jul 2025

Spartan Racing, Formula SAE | San Jose, CA

Python, Pygame, Matplotlib, PID control, vehicle dynamics, CSV logging

- Built a physics-based EV simulation to model acceleration from vehicle mass, drag, gearing, and motor torque
- Implemented PID-based power limiting to the **80 kW** rule and logged runs to CSV for repeatable tuning and validation

SR-Wireless-CAN, Vehicle Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, FastAPI, WebSockets, SQLite, Docker

- Proposed and implemented wireless firmware flashing for the VCU by decoding the vendor tool's CAN sequence from trace captures and reimplementing it in Python
- Developed CAN capture and diagnostics tooling with DBC-driven signal decoding for live vehicle testing and recorded trace replay